The complement clique number does not bound the Lovász theta function

Disproof of Conjecture 00000008908

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Abstract

For the five-cycle $G = C_5$, both its independence number and the clique number of its complement are two. An explicit rational Gram certificate for the actual Lovász theta semidefinite program gives $\vartheta(G) \geq 5041/2257 > 2$. Thus these two numbers cannot be the endpoints of the claimed monotone chain containing $\vartheta(G)$. Lean checks the full semidefinite feasible set and its bounded supremum, as well as the actual graph parameters.

The exact endpoint assertion

The English source explicitly names the endpoints as the independence number and the *clique number of the complement*. In particular its proposed sandwich would require

$$\alpha(G) \leq \vartheta(G) \leq \omega(\overline{G}).$$

The usual sandwich has a different upper endpoint: the chromatic number $\chi(\overline{G})$. The present counterexample addresses the endpoint actually written in the source.

Let G be the simple cycle on $\mathbb{Z}/5\mathbb{Z}$, with i adjacent to $i+1$ and $i-1$. Its largest independent sets have size two. Indeed, three independent vertices would require three cyclic gaps of at least two, impossible on five vertices; the pair $\{0, 2\}$ witnesses size two. Independent sets of G are precisely cliques of its complement, so

$$\alpha(G) = \omega(\overline{G}) = 2.$$

The Lean development proves these parameters using Mathlib's actual simple graph, complement, clique and independence definitions, with complete finite certificates.

The actual theta optimization problem

Use the standard equivalent semidefinite formulation [2, Theorem 4]:

$$\vartheta(G) = \sup \left\{ \sum_{i,j} X_{ij} : X = X^{\mathsf{T}} \succeq 0, \quad \operatorname{tr} X = 1, \quad X_{ij} = 0 \text{ if } i \sim_G j \right\}.$$

The supremum is over *all* feasible real matrices, rather than only the certificate constructed below.

An exact rational Gram certificate

Let B be the symmetric circulant matrix with first row $(-9, 32, 8, 8, 32)$, namely

$$B = \begin{pmatrix} -9 & 32 & 8 & 8 & 32 \\ 32 & -9 & 32 & 8 & 8 \\ 8 & 32 & -9 & 32 & 8 \\ 8 & 8 & 32 & -9 & 32 \\ 32 & 8 & 8 & 32 & -9 \end{pmatrix}.$$

Direct integer multiplication gives

$$B^{\mathsf{T}}B = \begin{pmatrix} 2257 & 0 & 1392 & 1392 & 0 \\ 0 & 2257 & 0 & 1392 & 1392 \\ 1392 & 0 & 2257 & 0 & 1392 \\ 1392 & 1392 & 0 & 2257 & 0 \\ 0 & 1392 & 1392 & 0 & 2257 \end{pmatrix}.$$

Set $X = B^{\mathsf{T}}B/11285$. It is positive semidefinite because, for every real vector z,

$$z^{\mathsf{T}}Xz = \frac{\|Bz\|_2^2}{11285} \geq 0.$$

Its trace is $5 \cdot 2257/11285 = 1$, and all positions corresponding to edges of C_5 are zero. Thus it is an actual feasible matrix. Its objective is

$$\sum_{i,j} X_{ij} = \frac{5 \cdot 2257 + 10 \cdot 1392}{11285} = \frac{5041}{2257} = 2 + \frac{527}{2257} > 2.$$

For completeness, the full feasible objective set is bounded. If $Y \succeq 0$ is symmetric, testing on $e_i - e_j$ gives $2Y_{ij} \leq Y_{ii} + Y_{jj}$. Summing over all 25 ordered pairs and using $\operatorname{tr} Y = 1$ yields

$$2\sum_{i,j} Y_{ij} \leq 10, \qquad \sum_{i,j} Y_{ij} \leq 5.$$

The certificate makes the feasible set nonempty. Hence its actual real supremum exists, and it is at least the certificate's objective. No empty-set or unbounded-set convention for the supremum is involved.

Contradiction and scope

Theorem 1. *For the actual simple five-cycle,*

$$\alpha(C_5) = \omega(\overline{C_5}) = 2 < \frac{5041}{2257} \leq \vartheta(C_5).$$

Therefore the claimed monotone chain with the two stated endpoints is false.

Proof. The graph parameters were computed above. The explicit matrix X is feasible for the full theta optimization problem and has objective strictly larger than two. □

The proof does not need the familiar exact value $\vartheta(C_5) = \sqrt{5}$, and does not claim to formalize that exact value or all versions of the sandwich theorem. It proves a sufficient strict rational lower bound against the exact upper endpoint named in the conjecture.

Formalization and reproducibility

The Lean graph is Mathlib's `cycleGraph 5`. `Feasible` uses the actual matrix predicate `PosSemidef`, trace one and all actual edge-zero conditions. `theta` is the real supremum of the objective values of every such matrix. The proof establishes nonemptiness, a bound for the whole feasible set, and the supremum inequality for the Gram certificate. The matrix entries and finite graph certificates are exact; no floating-point optimizer or untrusted numerical oracle is used.

Lean 4.19.0 and Mathlib revision
`c44e0c8ee63ca166450922a373c7409c5d26b00b` are pinned. From `lean/`, run

```
lake exe cache get
lake build
lake env lean -DwarningAsError=true Main.lean
```

The full project is rebuilt afresh and checked with warnings treated as errors. Printed theorem dependencies are audited for standard foundational axioms only. No proof placeholders or custom axioms are used. Compile the paper with `tectonic main.tex`.

The exact bilingual source, provenance, actual build logs, artifact hashes and all-page PDF review are included. An agent developed the proof and supplied a self-review; the parent reviewed the complete proof and final artifacts separately. No external independent review or novelty of this standard theta counterexample is claimed.

References

[1] The Last Math Competition, Conjecture 00000008908. Exact bilingual source and provenance included.

[2] L. Lovász, *On the Shannon Capacity of a Graph*, IEEE Transactions on Information Theory, 25(1), 1–7 (1979), Theorem 4. Original-paper mirror. This supplies the standard theta definition; the full certificate needed for this disproof is displayed above.